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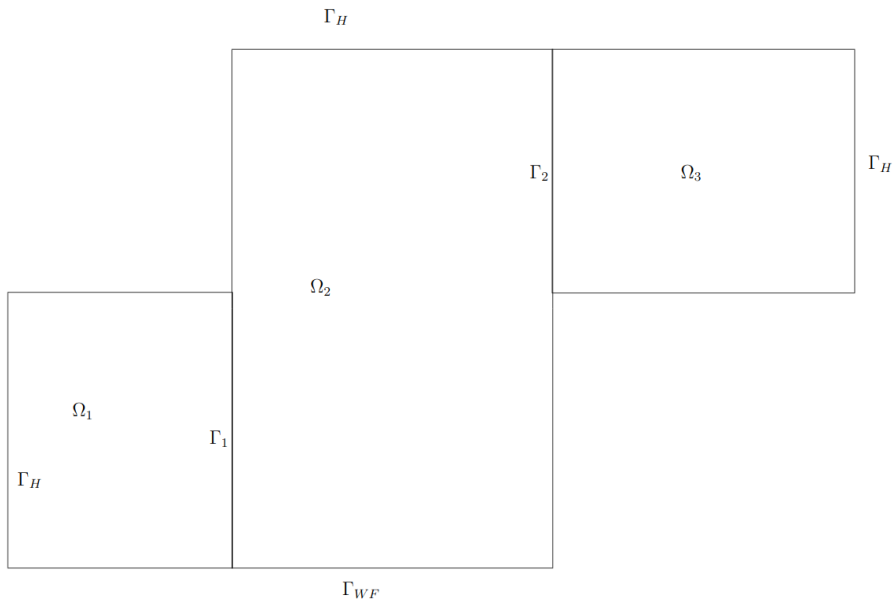
Advanced Numerical Algorithms with Python



The problem

Discretization

Parallelization



The problem

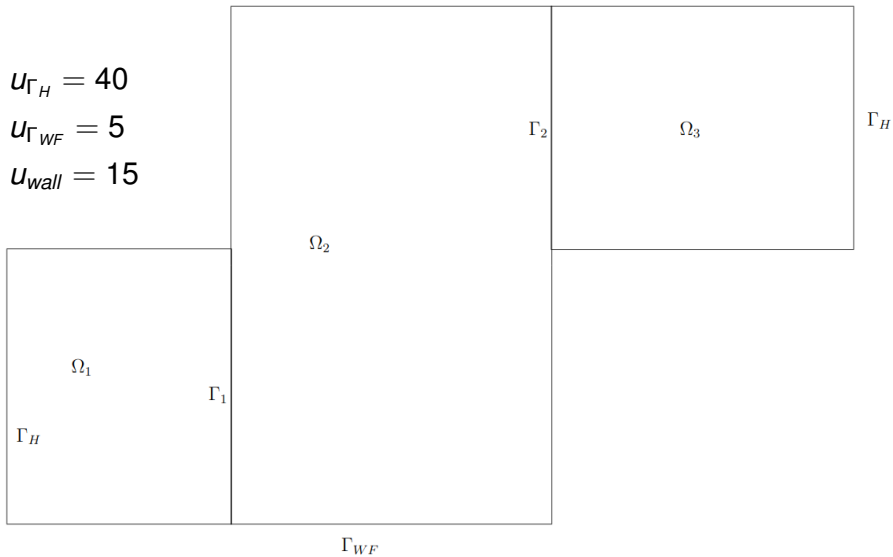
Discretization

Parallelization

$$u_{\Gamma_H} = 40$$

$$u_{\Gamma_{WF}} = 5$$

$$u_{wall} = 15$$



Laplace equation

The problem

Discretization

Parallelization

$$\Delta u(\mathbf{x}) = u_{xx} + u_{yy} = 0, \quad \mathbf{x} = (x, y) \in \Omega \subset \mathbb{R}^2$$



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Laplace equation

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$$\Delta u(\mathbf{x}) = u_{xx} + u_{yy} = 0, \quad \mathbf{x} = (x, y) \in \Omega \subset \mathbb{R}^2$$

■ Discretize



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Laplace equation

The problem

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- Discretize
- Parallelize



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Laplace equation

The problem

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$$\Delta u(\mathbf{x}) = u_{xx} + u_{yy} = 0, \quad \mathbf{x} = (x, y) \in \Omega \subset \mathbb{R}^2$$

- Discretize
- Parallelize
- Solve



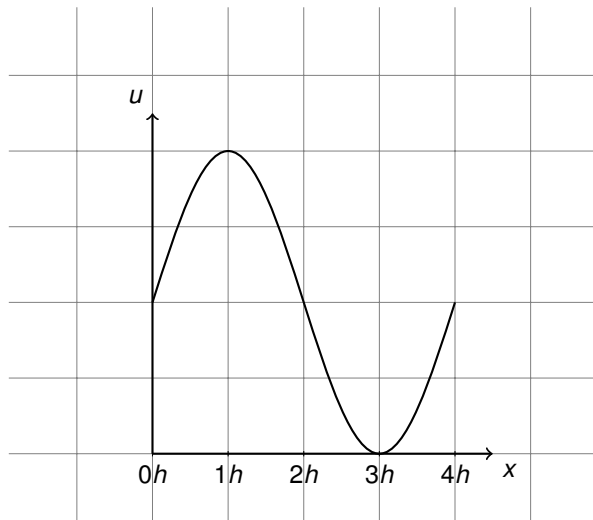
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Discretization: 1D

The problem

Discretization

Parallelization



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Discretization: Taylor expansion

The problem

Discretization

Parallelization

Taylor expand $u(x_{i+1})$ around $u(x_i)$:

$$u(x_{i+1}) = u(x_i) + hu'(x_i) + \frac{h^2}{2}u''(x_i) + \dots$$



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Discretization: Taylor expansion

The problem

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$$u(x_{i+1}) = u(x_i) + hu'(x_i) + \frac{h^2}{2}u''(x_i) + \dots$$

Solve for $u'(x_i)$:

$$u'(x_i) = \frac{u(x_{i+1}) - u(x_i)}{h} + \mathcal{O}(h)$$



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Discretization: Taylor expansion

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$$u'(x_i) = \frac{u(x_{i+1}) - u(x_i)}{h} + \mathcal{O}(h)$$

Choose discretization:

$$v'_i = \frac{v_{i+1} - v_i}{h}$$

Here, v denotes the discrete solution variable.



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2nd derivative

The problem

Discretization

Parallelization

Use the finite difference approximation applied the first derivative in order to obtain an approximation for the second derivative:

$$v_i'' = \frac{v_i' - v_{i-1}'}{h} = \frac{\frac{v_{i+1} - v_i}{h} - \frac{v_i - v_{i-1}}{h}}{h} = \frac{v_{i+1} - 2v_i + v_{i-1}}{h^2}$$



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2nd derivative

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Exercise: What is the accuracy of the approximation?



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Discretization: 2D

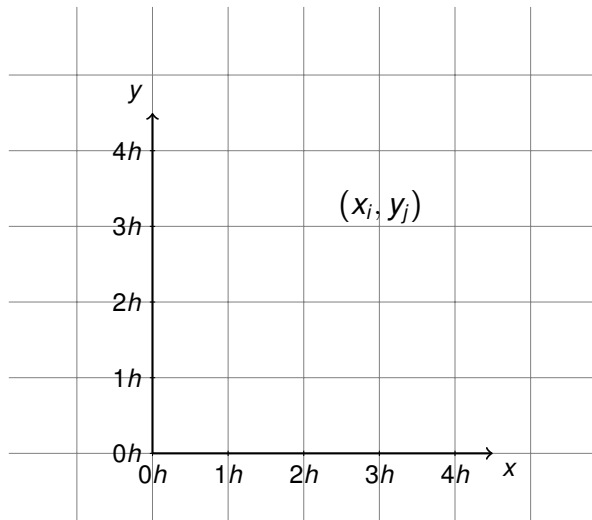
The problem

Discretization

Parallelization



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Discretization: 2D

The problem

Discretization

Parallelization

Simplifying assumption: Use grid spacing h in both dimensions.

Repeating the same exercise in x and y direction gives approximation

$$\frac{v_{i+1,j} + v_{i-1,j} - 4v_{i,j} + v_{i,j+1} + v_{i,j-1}}{h^2}$$



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Discretization: 2D

The problem

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Parallelization

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$$\frac{v_{i+1,j} + v_{i-1,j} - 4v_{i,j} + v_{i,j+1} + v_{i,j-1}}{h^2}$$

Collect in a matrix: $\Delta u \approx A\mathbf{v}$

$$A = \frac{1}{h^2} \begin{bmatrix} & & & \ddots & & & & & \\ & 1 & \dots 0 & \dots & 1 & -4 & 1 & \dots 0 & \dots & 1 \\ & & 1 & \dots 0 & \dots & 1 & -4 & 1 & \dots 0 & \dots & 1 \\ & & & 1 & \dots 0 & \dots & 1 & -4 & 1 & \dots 0 & \dots & 1 \\ & & & & \ddots & & & & & & \\ & & & & & \ddots & & & & & \end{bmatrix}$$



Discretization: 2D

The problem

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Parallelization

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Exercise: What changes if $h_x \neq h_y$?

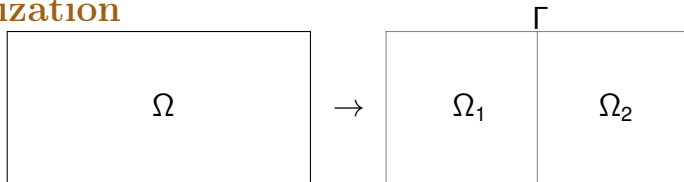


Parallelization

The problem

Discretization

Parallelization



- We could solve the problem on a single processor (left)
→ Slow if h is small.



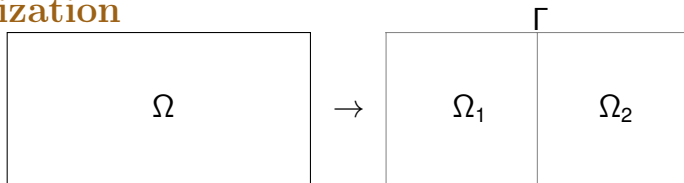
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Parallelization

The problem

Discretization

Parallelization



- We could solve the problem on a single processor (left)
→ Slow if h is small.
- Use two (or more) processors (right)
→ **Domain decomposition**



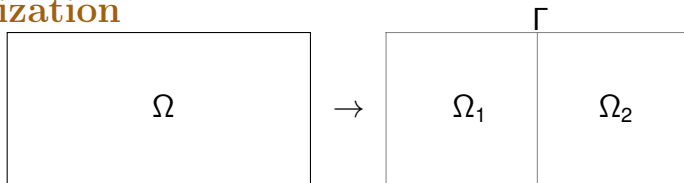
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Parallelization

The problem

Discretization

Parallelization



- We could solve the problem on a single processor (left)
→ Slow if h is small.
- Use two (or more) processors (right)
→ **Domain decomposition**
- The interface Γ is artificial
→ No information available from the other side of Γ
→ The problem has fundamentally changed



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Dirichlet-Neumann Iteration

The problem

Discretization

Parallelization

- We need a method to solve the decomposed problem such that the solution converges to that of the original problem.



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Dirichlet-Neumann Iteration

The problem

Discretization

Parallelization

- We need a method to solve the decomposed problem such that the solution converges to that of the original problem.
- Communication between processor 1 and 2 should only happen through Γ .



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Dirichlet-Neumann Iteration

The problem

Discretization

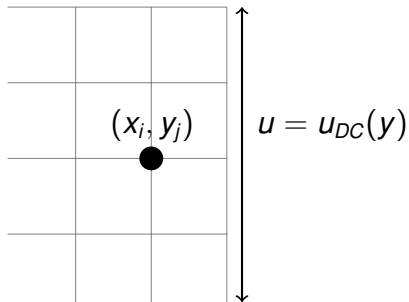
Parallelization

- We need a method to solve the decomposed problem such that the solution converges to that of the original problem.
- Communication between processor 1 and 2 should only happen through Γ .
- Solution: Alternate between using Dirichlet and Neumann conditions.



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Dirichlet condition



$$\Delta u(x_i, y_j) \approx \frac{v_{i,j-1} + v_{i,j+1} - 4v_{i,j} + v_{i-1,j} + u_{DC}(y_j)}{h^2} = 0$$

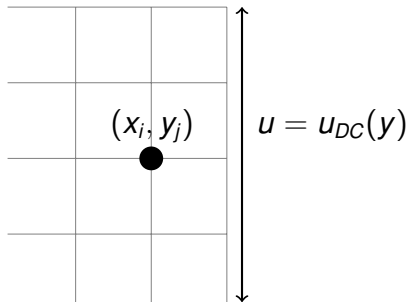


Dirichlet condition

The problem

Discretization

Parallelization



$$\Delta u(x_i, y_j) \approx \frac{v_{i,j-1} + v_{i,j+1} - 4v_{i,j} + v_{i-1,j} + u_{DC}(y_j)}{h^2} = 0$$

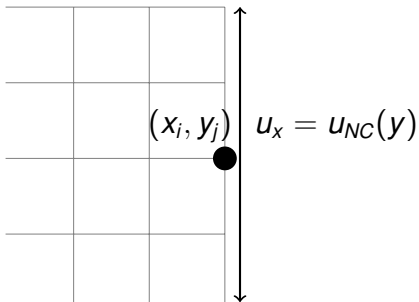
Move data over to right-hand side:

$$\frac{v_{i,j-1} + v_{i,j+1} - 4v_{i,j} + v_{i-1,j}}{h^2} = -\frac{u_{DC}(y_j)}{h^2}$$



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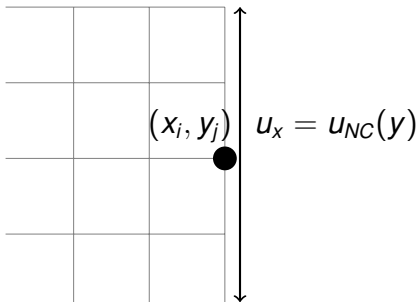
Neumann condition



$$\Delta u(x_i, y_j) \approx \frac{\frac{v_{i,j+1} - v_{i,j}}{h} - \frac{v_{i,j} - v_{i,j-1}}{h} + u_{NC}(y_j) - \frac{v_{i,j} - v_{i-1,j}}{h}}{h} = 0$$



Neumann condition



$$\Delta u(x_i, y_j) \approx \frac{\frac{v_{i,j+1} - v_{i,j}}{h} - \frac{v_{i,j} - v_{i,j-1}}{h} + u_{NC}(y_j) - \frac{v_{i,j} - v_{i-1,j}}{h}}{h} = 0$$

Move data over to right-hand side:

$$\frac{v_{i,j+1} + v_{i,j-1} - 3v_{i,j} + v_{i-1,j}}{h^2} = -\frac{u_{NC}(y_j)}{h}$$

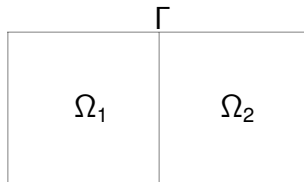


Dirichlet-Neumann Iteration

The problem

Discretization

Parallelization



- Step 1: On Ω_2 - Obtain $\mathbf{v}_\Gamma^k$. Send to Ω_1 .
- Step 2: On Ω_1 - Receive $\mathbf{v}_\Gamma^k$ from Ω_2 . Solve left system with $\mathbf{v}_\Gamma^k$ as Dirichlet condition. From $\mathbf{v}_{\Omega_1}^{k+1}$, compute Neumann condition at Γ . Send to Ω_2 .
- Step 3: On Ω_2 - Receive data from Ω_1 . Solve right system with Neumann condition. Obtain $\mathbf{v}_{\Omega_2}^{k+1}$.
- Step 4: Relax - $\mathbf{v}_{\Omega_{1,2}}^{k+1} \leftarrow \omega \mathbf{v}_{\Omega_{1,2}}^{k+1} + (1 - \omega) \mathbf{v}_{\Omega_{1,2}}^k$.



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